



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 • STANDARD 6.NOS.1

# Divide Whole Numbers by Fractions

Lesson 6-9 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 3 Enrichment



### TODAY'S OBJECTIVES

**Content Objective:** I can divide a whole number by a fraction by multiplying by the reciprocal.

**Language Objective:** I can explain my steps using the words fraction, reciprocal, quotient, and Keep, Change, Flip.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Whole Number

Fraction

Reciprocal

Keep, Change, Flip

Quotient



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A counting number like 0, 1, 2, or 3 with no fraction or decimal part is a

2

A number that shows part of a whole, like  $\frac{3}{4}$ , is a .

3

The fraction you get by flipping a fraction upside down is its

4

To divide by a fraction, I  : keep the first number, change  $\div$  to  $\times$ , and flip the second fraction.

5

The answer to a division problem is the .

## Write About the Math

**How does dividing a whole number by a fraction help the detective?**

*¿Cómo ayuda dividir un número entero entre una fracción al detective?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Whole Number**  
*Número entero*

☐ **Fraction**  
*Fracción*

☐ **Reciprocal**  
*Recíproco*

☐ **Keep, Change, Flip**  
*Mantener, cambiar,  
invertir*

☐ **Quotient**  
*Cociente*

**Model:** Each whole pound makes 2 half-pound bags, so 4 pounds makes 8 bags. — Students see that  $4 \div 1/2 = 8$  and explain there are 2 halves in each whole pound.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The detective can set up \_\_\_\_ markers because  $10 \div 1/4 =$  \_\_\_\_.**
- **My answer is \_\_\_\_ because \_\_\_\_.**

*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My evidence is \_\_\_\_, so \_\_\_\_.**  
*Mi evidencia es \_\_\_\_, entonces \_\_\_\_.*

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### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Whole Number
2. **Notes 2:** Fraction
3. **Notes 3:** Reciprocal
4. **Notes 4:** Keep, Change, Flip
5. **Notes 5:** Quotient
6. **Watch for this mistake:** *A common mistake in Divide Whole Numbers by Fractions is doing 'Keep, Change' but skipping the 'Flip' — students multiply by the divisor as it is instead of by its reciprocal. For example, for  $6 \div 3/4$ , a student might write  $6 \times 3/4 = 4\ 1/2$  and stop there, but the divisor  $3/4$  must be flipped to  $4/3$  first:  $6 \times 4/3 = 24/3 = 8$ . Before you submit, check: did I flip the second fraction (the divisor) upside down before multiplying?*
7. **Try It 1:** A.  $4\ 1/2$  pouches (4 full, half a pouch left) —  $3 \div 2/3 = 3 \times 3/2 = 9/2 = 4\ 1/2$ . You can fill 4 full pouches with half a pouch of mix left over.
8. **Try It 2:** A.  $12 - 9 \div 3/4 = 9 \times 4/3 = 36/3 = 12$  pans.